Richard C. Liu

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Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science in Computer Engineering *Sept. 2025 – Present*

Coursework: Programming in **C++**, Data structures and Algorithms, Digital Computers, Materials Chemistry, Numerical Methods, Advanced Calculus 1, Electronic Circuits 1, Electricity and Magnetism, Digital Circuits and Systems, Linear Circuits, Discrete Math and Logic, Calculus I/II, Linear Algebra, Project Studio, Mechanics

Experience

PaymentEvolution May 2026 – Aug. 2026
Software Developer Intern

- Built a **dual-authentication** payroll timesheet feature combining **facial recognition** and **PIN verification**; implemented through **YOLOv8** face detection, **ArcFace** embeddings, **FastAPI** endpoints, **PostgreSQL/pgvector** cosine search, **API rate limiting**, and secure **PIN hashing**.
- Deployed an automated insurance quote-intelligence pipeline where **Microsoft Graph push notifications** detected new Outlook emails, a **Cloudflare Worker/FastAPI webhook** received events and cached request metadata in **Redis** where **Celery** workers fetched email bodies and attachments through the **Outlook REST API**.
- Ingested email body and attachment metadata stored locally, corresponding directory metadata in **PostgreSQL** using **SQLModel** and **Alembic**. Data was fed into a **FastAPI LLM pipeline** that generated side-by-side vendor comparisons and carrier recommendations, processing **500+ emails/week** unattended and reducing **2 hours** of manual spreadsheet work.

Carleton University Feb. 2026 – Present
Student Researcher

- Completed graduate-level **Hyperdimensional Cognitive Models** coursework as an undergraduate; used **Holographic Reduced Representations** and **NumPy** to model Raven's Progressive Matrices reasoning tasks.

University of Waterloo Formula Electric Sept 2025 - May 2026
Firmware and Electrical Team Member

- Developed low-level **C firmware** drivers for **UART** Hardware-in-the-Loop data transmission and embedded control logic for a **DAC6551-Q1** voltage-output module.

Projects

Jiu-Jitsu Computer Vision Scoring System
Python, FastAPI, React, PyTorch, Supabase, AWS S3

- Built a **computer vision** web app that analyzes Brazilian Jiu-Jitsu footage through **pose estimation**, **multi-object tracking**, feature engineering, classification, scoring, and video annotation with **REST** and **WebSocket APIs**.
- **YOLO11-Pose** with a custom **BoT-SORT** tracker, reducing track-ID fragmentation from **17–29** to **5–10** and a **PyTorch** MLP classifier on **47.8K samples** to **96.2% test accuracy** to compute positional point scoring.
- Integrated a **Gemini VLM** with schema-constrained **JSON outputs**; used **Supabase** with Postgres to store video-processing job status, buckets for uploaded and output videos; **AWS S3** for MLP weights, **Modal GPU** for MLP processing, **Alembic**, **Docker**, and rate-limited APIs for scalable video processing for submission recognition.

Full-Stack Barbershop Booking Web App
React, Vite, Express, TypeScript, PostgreSQL, Prisma

- Built a **React 19/Vite** SPA and **Express/TypeScript REST API** with customer accounts, service browsing, live slot availability, admin scheduling, **JWT authentication**, and **bcrypt** password hashing.
- Modeled **PostgreSQL** data with **Prisma ORM**, enforced double-booking prevention through unique constraints, added **Nodemailer SMTP** notifications, and secured deployment with **Helmet**, **rate limiting**, **Zod validation**, input sanitization, **CORS**, and **Vercel CI/CD**.

Network-Connected Embedded MP3 Player
C/C++, Wi-Fi, REST, OAuth, I2C, Embedded Systems

- Implemented interrupt-driven rotary encoder decoding with a **Gray code state machine** and debouncing for **95%+ accuracy**; built **Wi-Fi** uploads, a **RESTful web server**, **Spotify Web API** playback control, **OAuth 2.0**, JSON parsing, and an **I2C OLED UI**.

Skills

Languages: C, C++, Python, JavaScript, TypeScript, Java, SQL
Frontend: React, Vite, Next.js, HTML5, CSS3, Tailwind CSS
Backend/Data: FastAPI, Node.js, Express, REST APIs, Webhooks, WebSockets, PostgreSQL, Redis, Celery, Prisma, SQLModel, Alembic, pgvector
AI/ML: Prompt Engineering, LLM Pipelines, Claude, Gemini, VLMs, Computer Vision, YOLO, PyTorch, NumPy, Embeddings, Vector Search
Cloud/DevOps: Docker, Cloudflare Workers, AWS S3, Supabase, Vercel, Modal, CI/CD, Git, GitHub, Linux
Embedded/Engineering: Embedded C, UART, I2C, Verilog, Arduino, PlatformIO, Altium Designer, STM32CubeIDE, Soldering, System Design, API Design